



A.5 Galilean and special relativity (HL)

Space, time and motion · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Galilean relativity assumes space and time are absolute and velocities add simply ($u' = u - v$) between inertial reference frames. Einstein's special relativity instead postulates that the laws of physics are the same in all inertial frames and the speed of light in vacuum, c , is the same for all observers regardless of relative motion; this forces time dilation, $\Delta t = \gamma \Delta t_0$, and length contraction, $L = L_0/\gamma$, where $\gamma = 1/\sqrt{1 - v^2/c^2}$ and Δt_0 , L_0 are proper time and proper length measured in the rest frame. Relativistic mass-energy equivalence gives $E = \gamma mc^2$, with rest energy $E_0 = mc^2$ and total energy including kinetic energy $E_k = (\gamma - 1)mc^2$.

Worked example: follow the reasoning

A muon has a proper lifetime of $2.2 \mu\text{s}$ in its own rest frame. It travels at $v = 0.98c$ relative to Earth. Find the lifetime measured by an observer on Earth.

$$\gamma = 1/\sqrt{1 - 0.98^2} = 1/\sqrt{1 - 0.9604} = 1/\sqrt{0.0396} = 1/0.1990 = 5.03.$$

$$\Delta t = \gamma \Delta t_0 = 5.03 \times 2.2 \mu\text{s} = 11.1 \mu\text{s}.$$

Answer: $\Delta t \approx 11 \mu\text{s}$.

Quick check · multiple choice

A spaceship moves at $v = 0.80c$ relative to Earth. What is the Lorentz factor γ (to 2 s.f.)?

- A 0.60
- B 1.0
- C 1.3
- D 1.7

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A spacecraft of proper length 120 m travels past Earth at $v = 0.60c$. [6]
- (a) Calculate the Lorentz factor γ for this speed.
 - (b) Calculate the length of the spacecraft as measured by an observer on Earth.
 - (c) The spacecraft crew measures a proper time interval of 10.0 s for an event onboard. Calculate the corresponding time interval measured on Earth.

Final answer: _____



Answers and explanation

Question 1 [6]

(a) $\gamma = 1/\sqrt{1-0.36} = 1/\sqrt{0.64} = 1/0.80 = 1.25$.

(b) $L = L_0/\gamma = 120/1.25 = 96.0 \text{ m}$.

(c) $\Delta t = \gamma \Delta t_0 = 1.25 \times 10.0 = 12.5 \text{ s}$.

Quick check answer

D. $\gamma = 1/\sqrt{1-0.64} = 1/\sqrt{0.36} = 1/0.60 = 1.67 \approx 1.7$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Physics HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.